

Visual Repair Report

This pass fixes Batch 1 and Batch 2 visual/text alignment issues, centralizes the canonical prompt-response example, and refreshes review/PDF artifacts for mobile-first human review.

What Changed

- Added `src/data/canonicalExamples.ts` as the shared source for the prompt, generated response, response-so-far, next-token choices, and attention-only variant.
- Updated Prompt vs Response, Tokenization, Token IDs, Embeddings, Vectors, Tensors, Autoregression, Prompt Run, exercises, glossary examples, and review cards to use the canonical example.
- Repaired visual aids so in-canvas labels are short and full explanations live in HTML captions and numbered callouts.
- Updated export paths to `docs/review/prompt-life-lesson-cards-v0-9-2.pdf` and `docs/content-inventory/prompt-life-lesson-cards-v0-9-2-visual-repair.pdf`.
- Bumped the visible app version to `v0.9.2`.

Canonical Example

User prompt: Give me one sentence with two household pets in conflict.

Generated response: A jealous dog chased a startled cat across the kitchen floor.

Response so far: A jealous dog chased a startled cat across the kitchen

Next-token candidates: floor, room, tiles, counter, quantum, elephant. Chosen token: floor.

Attention-only variant: A jealous dog chased a startled cat because it hissed. Teaching target: it -> cat.

Verification

- `npm install` : passed, 0 vulnerabilities.
- `npm run typecheck` : passed.
- `npm run build` : passed.
- `npm run build:pages` : passed.
- `npm run export:lesson-pdf` : passed.
- `npm run export:lesson-cards` : passed.
- Mobile visual-card overflow checks at 320px, 390px, and 430px: passed for the repaired visual cards.
- In-app browser review-route QA found the repaired visual cards and canonical text.

Files Changed

- `README.md`, `package.json`, and `scripts/export-lesson-pdf.mjs`
- `src/components/ConceptAnimations.tsx`, `src/components/ExerciseSystem.tsx`, and `src/components/VisualAids.tsx`
- `src/data/canonicalExamples.ts`, `src/data/content.ts`, `src/data/contentReview.js`, `src/data/exercises.ts`, and `src/data/promptRun.ts`
- `src/main.tsx` and `src/styles/global.css`
- v0.9.2 curriculum docs, lesson PDFs, screenshots, and this report PDF

How To Run Locally

- `cd ~/Desktop/promptlife`
- `npm install`
- `npm run dev`
- Open the Vite URL, usually `http://localhost:5173/`.
- For phone testing from GitHub Pages, use `https://kevinhegg.github.io/promptlife/` after `main` deploys.

Screenshots

TRAINING-LOOP

Training Loop



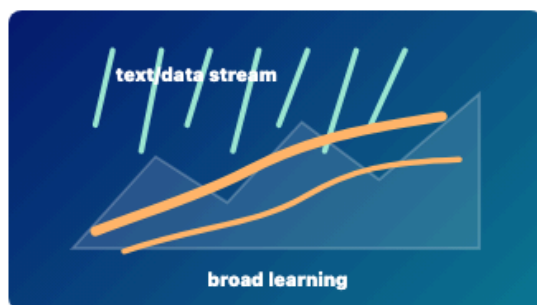
Training Loop

Predict, compare, loss, update weights, repeat. Training changes weights.

- 1 The model predicts a target.
- 2 The prediction is compared with the target.
- 3 Loss measures error.
- 4 Update weights is the durable-change step.
- 5 The loop repeats many times.

PRETRAINING-RAIN

Broad Pretraining



Broad Pretraining

A wide stream of data shapes broad durable learning before normal use.

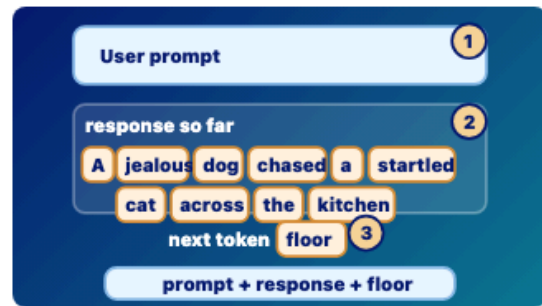
- 1 Many examples flow through the training loop.
- 2 Repeated updates shape broad model capability.

OVERFITTING-GENERALIZATION

Overfitting vs Generalization

PROMPT-RESPONSE

Prompt vs Response



Prompt vs Response

A complete user prompt is given; response tokens are generated and appended.

- 1 User prompt: Give me one sentence with two household pets in conflict.
- 2 Response so far: A jealous dog chased a startled cat across the kitchen.
- 3 Next token: floor. Next context combines prompt + response so far + floor.

TOKENIZATION

Text to Tokens



Text to Tokens

Generated response text is split into model-readable chunks before embedding lookup.

- 1 Generated response: A jealous dog chased a startled cat across the kitchen floor.
- 2 Simplified for learning. Real tokenizers may split differently.

TOKEN-IDS

Token IDs

Training Loop: clean five-step loop with durable update step.

Prompt vs Response: complete prompt, response so far, and next token.

TOKENIZATION

Text to Tokens



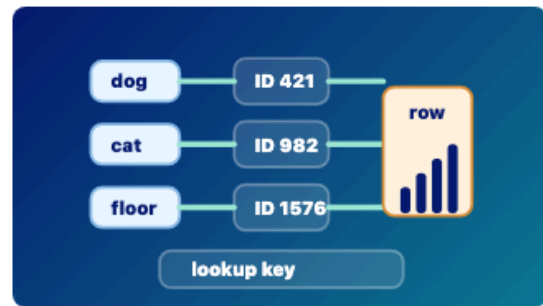
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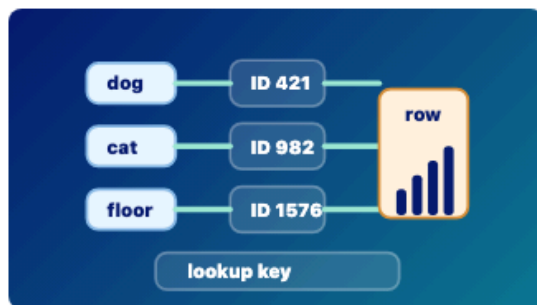
Token IDs

Each token gets a lookup number that points to an embedding row.

- 1 dog -> 421, cat -> 982, floor -> 1576.
- 2 The ID is a lookup key, not the meaning.
- 3 The ID points to an embedding row.

TOKEN-IDS

Token IDs



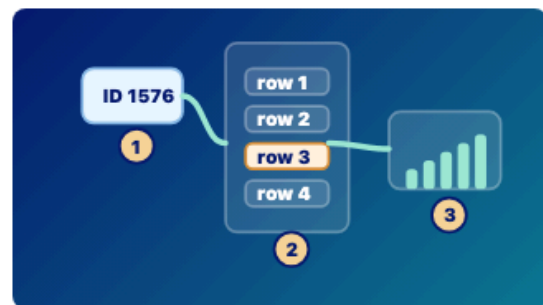
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EMBEDDINGS

Embedding Lookup



Embedding Lookup

A token ID retrieves a learned starting vector.

- 1 The embedding table was learned during training.
- 2 Inference retrieves one row temporarily for the current context.
- 3 The retrieved embedding is a starting vector, not a definition.

EMBEDDINGS

Embedding Lookup



VECTORS

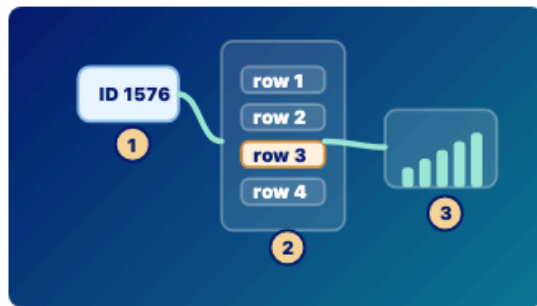
Feature Vector

Tokenization: canonical generated response split into simplified tokens.

Token IDs: dog, cat, and floor IDs shown as lookup keys.

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VECTORS

Feature Vector



Feature Vector

A vector is a list of numerical features, not a sentence.

- 1 Labels such as animal-ish, grammar role, tone, position, and context clue are simplified.
- 2 Real features are distributed across many dimensions.

VECTORS

Feature Vector



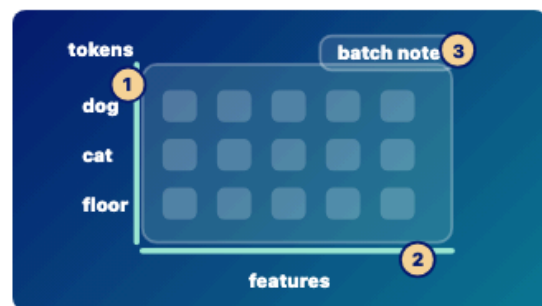
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TENSORS

Tensor Block



Tensor Block

Tensors organize many token vectors for layer-by-layer processing.

- 1 Token positions run down one axis.
- 2 Feature dimensions run across another axis.
- 3 Batch x tokens x features is an advanced shape note.

TENSORS

Tensor Block

ATTENTION

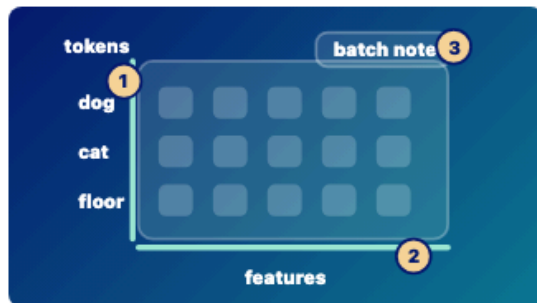
Attention Weave

Embeddings: ID lookup to table row to vector.

Vectors: simplified feature sliders plus distributed-feature note.

TENSORS

Tensor Block



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ATTENTION

Attention Weave



Attention Weave

Attention is weighted relevance between token positions.

MLP

MLP Reshape



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PROMPT ARRIVES

Prompt vs Response

Given context versus generated tokens

Stage: response-generation

Priority: low

Rubric avg: 4.4/5



Prompt vs Response

A complete user prompt is given; response tokens are generated and appended.

- 1 User prompt: Give me one sentence with two household pets in conflict.
- 2 Response so far: A jealous dog chased a startled cat across the kitchen.
- 3 Next token: floor. Next context combines prompt + response so far + floor.

DEFINITION

Prompt tokens are given to the model; response tokens are generated by the model one at a time and appended to the context.

CORE EXPLANATION

The user prompt should be a complete request, such as: "Give me one sentence with two household pets in conflict." The model then begins generating a response. During generation, the response-so-far may be incomplete: "A jealous dog chased a startled cat across the kitchen". After the model chooses one response token, such as "floor", that token is appended. The next forward pass sees the user prompt plus the response so far.

WHERE IT HAPPENS

During inference and response generation.



Tensors: stable 2D tokens x features grid.

DURABLE VS TEMPORARY

Prompt and response-so-far tokens are temporary context. They can influence the current run, but they do not normally update weights.

Print-route sample used by the lesson-card PDF exporter.

Known Remaining Issues

- Source review is still needed before publication-final claims about transformer internals.
- The tensor visual remains 2D for PDF reliability; any future 3D tensor needs a print-safe fallback.
- Real iPhone Safari testing remains the final post-deploy check.

Next Three v0.2 Improvements

- Implement Batch 3 as Attention, MLP, Layers, and Hidden States with the same canonical thread.
- Add source citations or bibliography notes for Batch 1 and Batch 2 concepts.
- Add a reusable screenshot/overflow script to make future visual-aid QA repeatable.